Lecture Notes
on Linear Regression
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1 Bivariate Data

1.1 Definition:

Bivariate data refers to data that involves **two variables** measured on the same individual, object, or event.

It helps in analyzing the relationship between the two variables, often using graphs, correlation, or regression methods.

- If the two variables are numerical, we usually plot them on a scatter diagram and study their relationship (positive, negative, or no correlation).
- If one variable depends on the other, we often explore this through **linear regression**.

1.2 General Representation of Bivariate Data:

Bivariate data with n observations (i.e., data points) can be represented as a set of ordered pairs:

$$\{(x_1, y_1), (x_2, y_2), (x_3, y_3), \ldots, (x_n, y_n)\},$$

(,) y-coordinate
x-coordinate.

where for $i = 1, 2, \ldots, n$,

- x_i is the value of the first variable (independent variable),
- y_i is the corresponding value of the second variable (dependent variable).

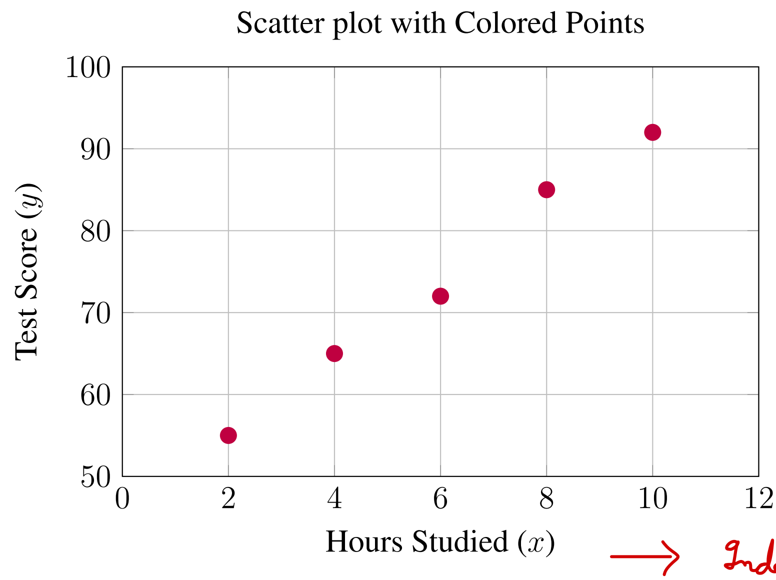
Tabular Representation: The same data can also be written in a two-column tabular form:

Variable 1: x_i	**Variable 2:** y_i
x_1	y_1
x_2	y_2
x_3	y_3
$\vdots$	$\vdots$
x_n	y_n

Example: Suppose we record the number of hours a student studies in a week (x) and their corresponding test score (y). The dataset below is an example of bivariate data:

Hours Studied (x)	Test Score (y)
2	55
4	65
6	72
8	85
10	92

Here, both variables are observed for the same students, and we can analyze whether an increase in study hours is associated with an increase in test scores.



dependent variable

Independent variable

2 Regression and Linear Regression

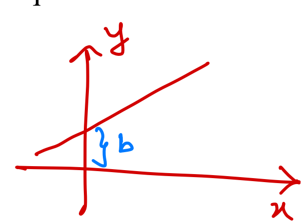
Definition of Regression: Regression is a statistical method used to study the relationship between a dependent variable and one or more independent variables.

It helps in predicting the value of the dependent variable based on the given values of the independent variables.

Definition of Linear Regression:
Linear regression is the simplest form of regression, where the relationship between the dependent variable y and the independent variable x is assumed to be linear, and is modeled by the equation:

$$y = mx + b,$$

where



- m is the slope of the line, indicating the rate of change of y with respect to x,
- b is the intercept, representing the value of y when $x = 0$.

Graphically, the data points are plotted on a scatter plot, and the straight line $y = mx + b$ is drawn to best describe the trend of the data.

Examples of Linear Relationship

☛ Example 1: Positive Linear Relationship

Consider the following dataset:

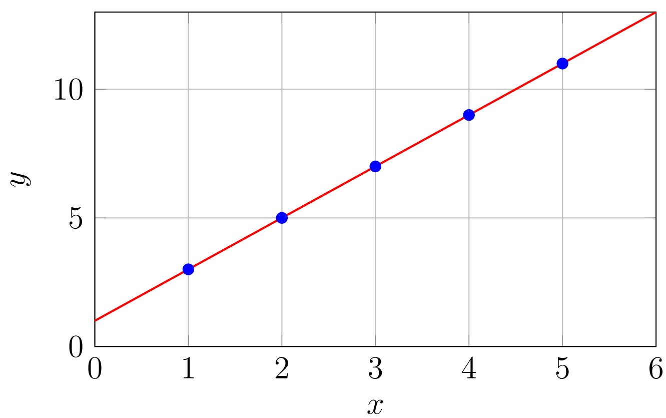
$$\{(1,3),(2,5),(3,7),(4,9),(5,11)\}$$

Here, the variables increase together. The best-fit line is exactly:

$$y = 2x + 1,$$

where the slope is $m = 2$ and the intercept is $b = 1$.

Positive Relationship



☛ Example 2: Negative Linear Relationship

Consider the dataset:

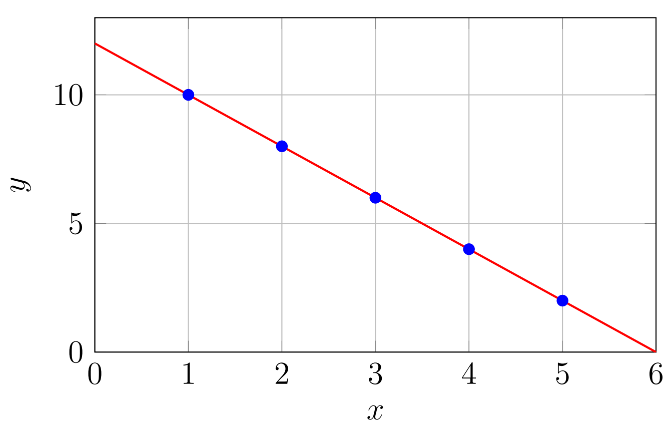
$$\{(1,10),(2,8),(3,6),(4,4),(5,2)\}$$

Here, as x increases, y decreases. The best-fit line is:

$$y = -2x + 12,$$

where the slope is $m = -2$ and the intercept is $b = 12$.

Negative Relationship



2.1 Slope of a Line Through Two Points

If a straight line passes through two distinct points

$$(x_1, y_1) \quad \text{and} \quad (x_2, y_2),$$

then the slope of the line is given by

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \qquad (x_1 \neq x_2).$$

Once the slope m is found, the equation of the line can be written using the point–slope form:

$$y - y_1 = m(x - x_1).$$

$\Rightarrow y - y_1 = mx - mx_1$

This can be rearranged into the slope–intercept form:

$\Rightarrow y = mx \underbrace{- mx_1 + y_1}_{}$

$y = mx + b$

$$y = mx + b,$$

$\therefore b = y_1 - mx.$

where

$$b = y_1 - mx_1.$$

$x_1 \; y_1 \quad x_2 \; y_2$

$(1,3), (2,5)$

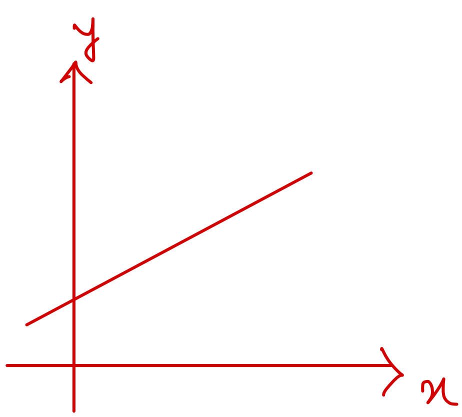
$m = \frac{5-3}{2-1} = 2$

$y - y_1 = m \cdot (x - x_1)$

$\Rightarrow y - 3 = 2(x - 1)$

$\Rightarrow y = 2x - 2 + 3$

$\Rightarrow y = 2x + 1$



$(1,10), (2,8).$

$10 - 8 = m(1-2)$

$2 = 1m - 2m$

$2 = -1m$

$m = -2$

$y - 10 = -2(x-1)$

$y - 10 = -2x + 2$

$y = -2x + 12$

2.2 Residuals in Linear Regression

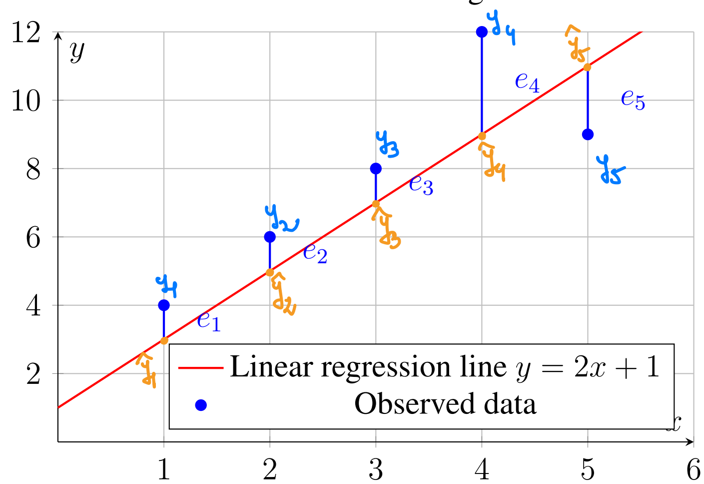
Definition: In regression analysis, a residual is the difference between the observed value of the dependent variable and the value predicted by the regression line. For a data point (x_i, y_i),

$$\text{Residual} = y_i - \hat{y}_i,$$

where $\hat{y}_i$ is the predicted value from the regression line. Residuals measure how far each data point lies from the fitted line.

Graphical Representation:

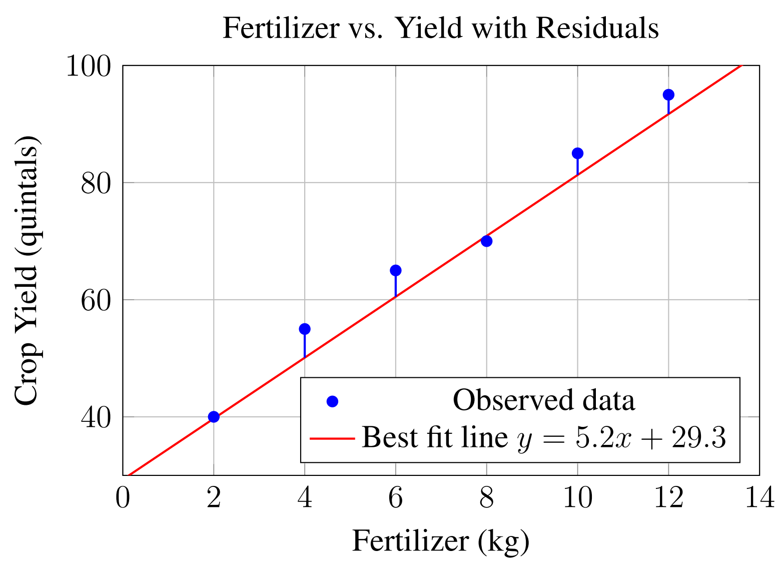
Residuals in Linear Regression



NOTE: When the line represents the linear regression line, the slope m and the y-intercept b are known as regression coefficients.

Example: A farmer wants to examine whether the amount of fertilizer used (x, in kg) affects the yield of crops (y, in quintals). The data collected from 6 plots of land is:

x (Fertilizer, kg)	y (Observed Yield)	$\hat{y}$ (Predicted Yield)	Residual $e_i = y - \hat{y}$
2	40	39.7	0.3
4	55	50.1	4.9
6	65	60.5	4.5
8	70	70.9	-0.9
10	85	81.3	3.7
12	95	91.7	3.3



Note on Residual Squares:
For each data point (x_i, y_i), the residual is

$$e_i = y_i - \hat{y}_i,$$

and the **residual square** is

$$e_i^2 = (y_i - \hat{y}_i)^2.$$

Adding these across all points gives the Residual Sum of Squares (R^2):

$$R^2 = \sum_{i=1}^{n} e_i^2.$$